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Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory's Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes. While this parameter provides a useful baseline for morphological classification, its binary nature prevents the optimal combination of information from multiple bands, as well as from other classification methods (e.g., those based on colors or external datasets). In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020). In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\rm PSF} - m_{\rm CModel}$ and magnitude $m_{\rm CModel}$. At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which, with appropriately chosen priors for the star-to-galaxy number ratio, allows us to compute p_S, the posterior probability that a source is unresolved. We apply this method to the Rubin DP1 and DP2 object catalogs and validate the resulting classifications using external HST labels in the COSMOS field. Using these labels as the ground truth, we quantitatively compare the performance of the p_S-based classifier with that of the Rubin `extendedness` parameter. We find that the p_S-based classifier consistently outperforms the Rubin `extendedness` baseline, with the improvement increasing as additional bands are incorporated. Approximately, the p_S classifier achieves the same classification performance at a magnitude limit about one magnitude fainter than the `extendedness` classifer. Depending on the scientific application, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

1. INTRODUCTION

The morphological classification of resolved and unresolved sources[4], commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin's Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies and the galaxies that do exist are obviously extended on the sky. Star-galaxy separation becomes increasingly harder at faint magnitudes, such as those probed by Rubin's Legacy Survey of Space

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[4] Note that some stars can be resolved, such as Asymptotic Giant Branch stars, while quasars and galaxies can be unresolved.

and Time (LSST; for details see Ž. Ivezić et al. 2019a and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller.

A variety of methods have been proposed in the literature for morphological classification of sources in images. C. T. Slater et al. (2020) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star-galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline (R. H. Lupton et al. 2002), or SExtractor pipeline (E. Bertin & S. Arnouts 1996), are closely related both to each other, and to the Bayesian model odds ratio, first derived by W. L. Sebok (1979). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best peformance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes, $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness=0`, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness=1`. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin's Data Preview 1 catalogs, see Fig. 28 in Vera C. Rubin Observatory Team et al. (2026).

Although `extendedness` was successfuly used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star-galaxy separation threshold is optimized at the faint, low S/N, end of the data, the ability to recognize barely resolved objects at the bright end is not fully exploited. More importantly, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands. With appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved), can be computed in Bayesian framwork.

The main goal of this paper is to implement the steps towards Bayesian star-galaxy separation outlined in SIL2020 using Rubin's Data Preview 2 data. We describe our dataset and method in §2, and quantify its performance using HST-based labels in §3. We discuss our results and how to retrieve our catalog with improved star-galaxy separation in §4.

2. METHODOLOGY AND DATA

Section outline

2.1. *Rubin Data Preview 2*

Intro to DP2

We choose COSMOS field because of HST labels J. R. Weaver et al. (2022)

XXX: Describe the mached sample: sky coverage and counts in Figure 1 (it will be split into 2 figures), what was the matching radius, matching fraction and total counts (DP2 all, HST stars, HST galaxies)?

2.2. *The distributions of sources classified using HST labels*

The top left panel in Figure 2 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\rm PSF} - m_{\rm CModel}$ vs. $m_{\rm CModel}$ diagram. Modeling this distribution is the key ingredient of the star-galaxy separation method developed here.

The top right panel in Figure 2 shows that galaxies, in addition to outnumbering stars at faint magnitudes, have on average bluer colors than stars.

Color-color diagrams shown in the bottom two panels in Figure 2 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 3 and the top left panel in Figure 2 show that increased noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods.

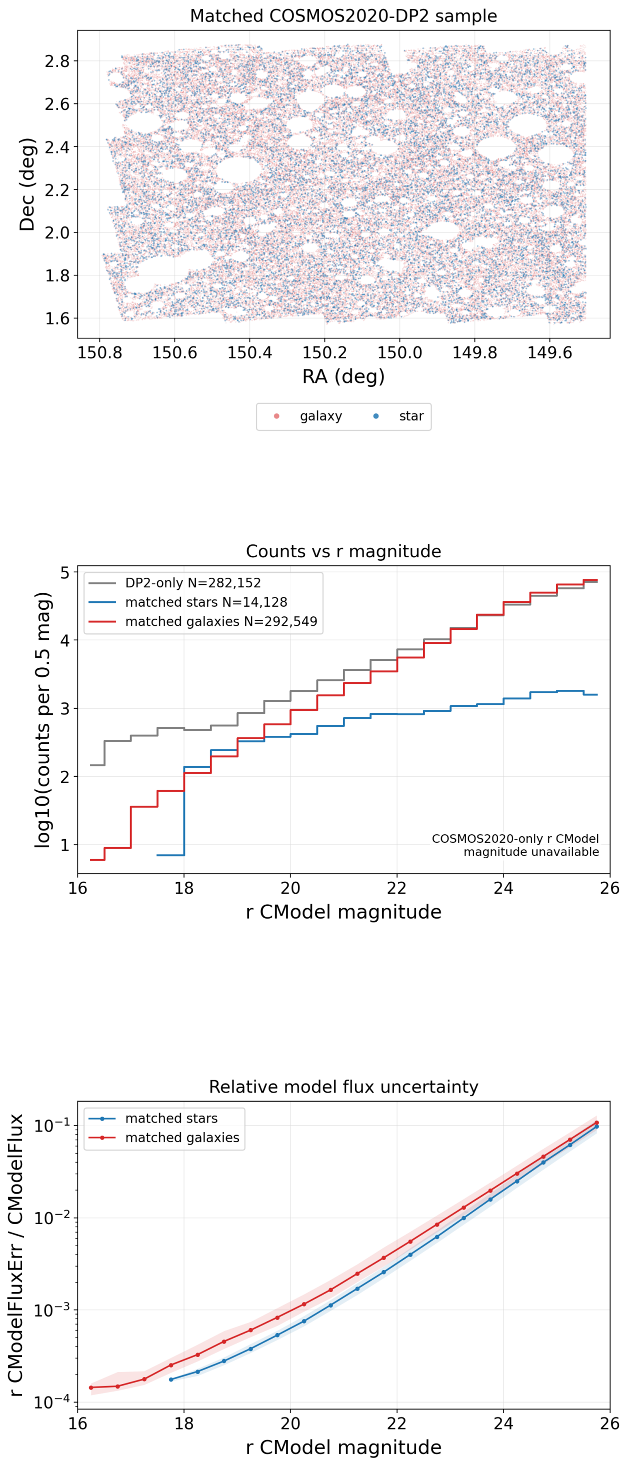

Figure 1. COSMOS/COSMOS2020 matched sample overview. The primary sample is selected using uncorrected r-band CModel magnitude, while colors are dust-corrected where applicable. XXX: SPLIT THIS FIGURE INTO 2 FIGURES; top panel: remove title; instead of galaxy and stars, show all DP2 sources in one color and overplot matches in other color; remove legend below (this information needs to go into caption; bottom panel: multiply counts by 2 to get unit to be per mag; remove title and text in the bottom right; HOW CAN RED LINE BE ABOVE GRAY LINE? It's probably some error in effective area or such... We don't need the third panel.

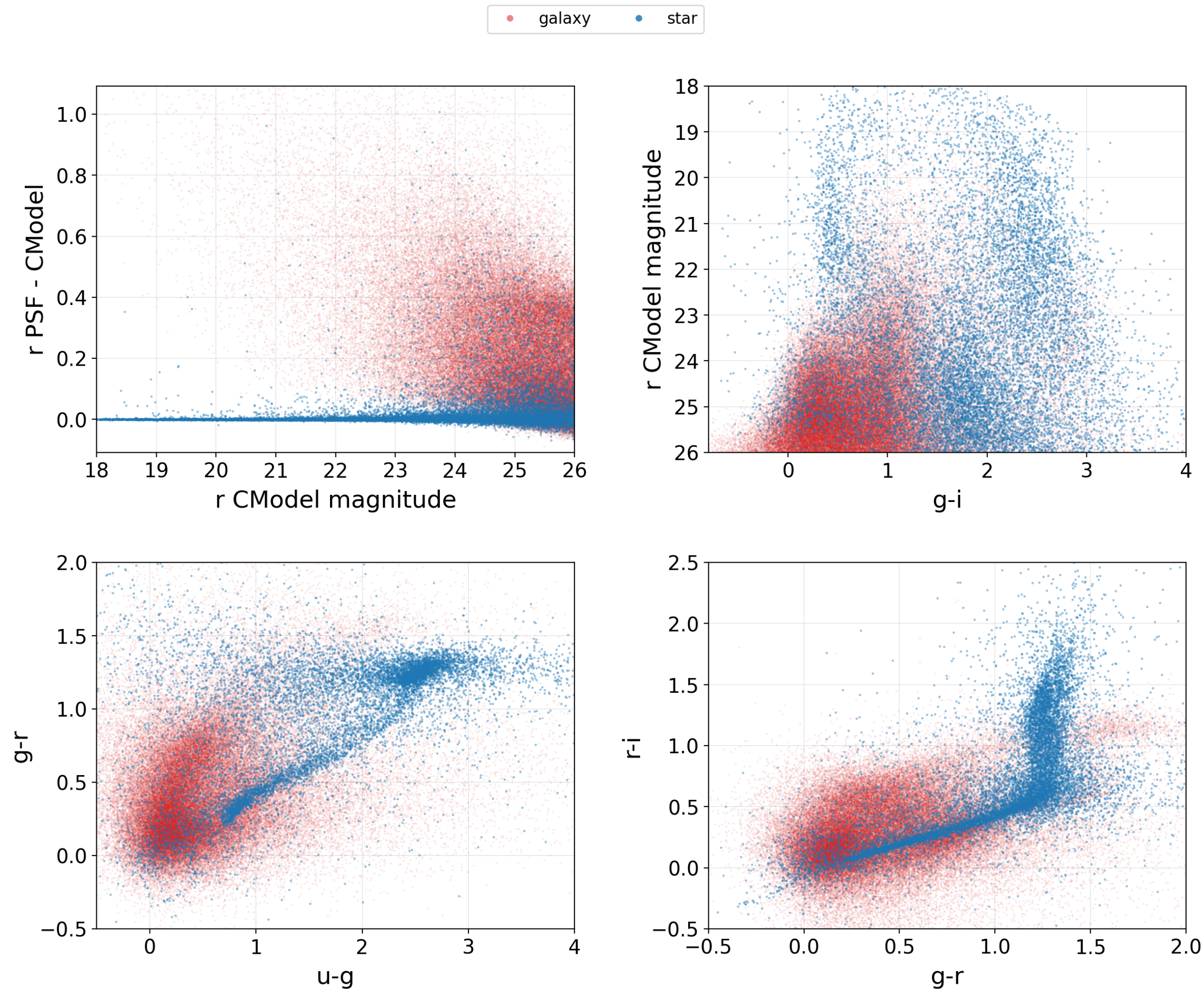

Figure 2. COSMOS/COSMOS2020 truth-label morphology and color overview. The r-band CModel magnitude is uncorrected for ISM dust extinction, while the CModel colors are dust-corrected where applicable. XXX: almost perfect! however, the counts plot shows the same numbers of stars and galaxies at about r = 19 (which is roughly correect) but the top right panel doesn't seem to be consistent with that - galaxies begin to dominate only at r of about 22-23. WE NEED TO DOUBLE CHECK THIS! Also, remove the title (here and elsewhere - the journal editors don't like it.

2.3. *The distributions of sources classified using* `extendedness` *parameter*

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 4, which compares subsamples classified using HST ground-truth labels and `extendedness` parameter in the r band. As evident, a significant fraction of faint sources within 2 magnitudes from the faint limit are misclassified when using default `extendedness` parameter.

2.4. *Bayesian Star-Galaxy Separation Method*

Summary of the method, expressions from SlL2020...

2.5. *Implementation of the SIL2020 method*

2.5.1. *Mixture Model for Deltam vs.* $m_{\rm CModel}$ *diagram*

XXX We need Nirav to describe in detail how the fits are done (what bins, what weights, etc). We also show figures that demonstrate performance (e.g. histograms and 2D residuals).

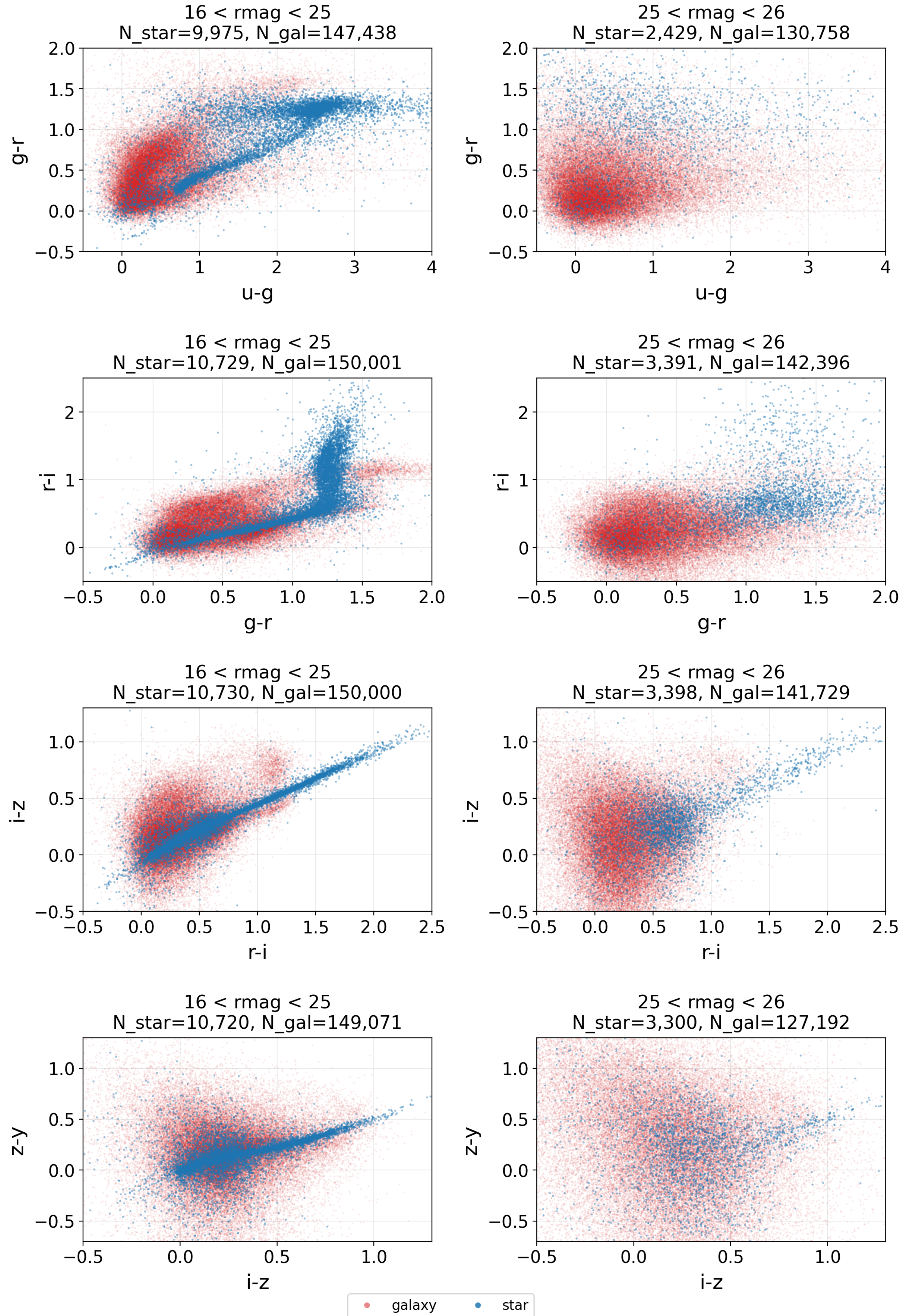

Figure 3. Similar to Figure 2, except that data are split by r-band CModel magnitude to show deterioration close to the faint magnitude limit (left: $16 < \mathrm{rmag} < 25$; right: $25 < \mathrm{rmag} < 26$).

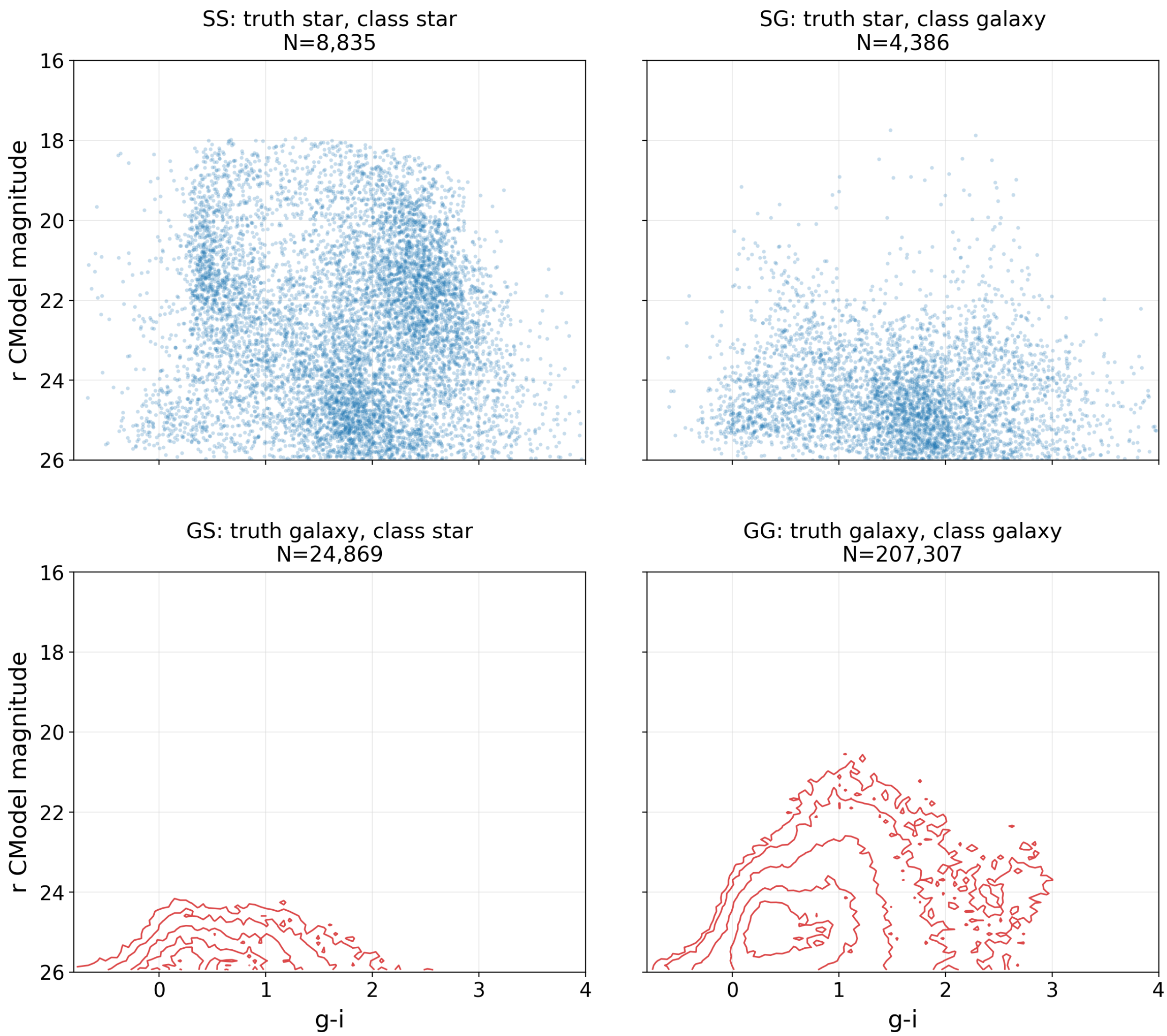

Figure 4. Performance of default `extendedness` parameter in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. g-i color-magnitude diagram, with counts shown in the title. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by `extendedness` is severely contaminated by galaxies for $m_{\rm CModel} > 24$.

2.5.2. *Priors for Star-Galaxy Counts ratio*

Here we discuss how the normalization parameters from the best-fit mixture can be used to set priors for the star-to-galaxy number ratio. We also compare to the HST curve, as well as to TRILEGAL-based curve.

3. PERFORMANCE OF THE BAYESIAN CLASSIFER

Present validation metrics, ROC curves, magnitude dependence, and comparison between extendedness and probabilistic classifiers...

4. DISCUSSION AND CONCLUSIONS

Reliable unresolved-source selection is important for Rubin science cases, including studies of Milky Way structure and stellar populations, quasar selection, photometric calibration, transient science, and the removal of stellar foregrounds from extragalactic samples. Maximizing the performance of star-galaxy separation is particularly for important for Galactic and stellar science cases, where, for example, searches for rare objects or low contrast stellar overdensities in the Galactic halo can be particularly hampered by the "noise" of misclassified galaxies contaminating the stellar sample.

Additional COLOR method!!

TODO: Discuss limitations, external-label coverage, faint-end behavior, and future improvements.

CONCLUSION: summarize the path toward robust Rubin star-galaxy separation.

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